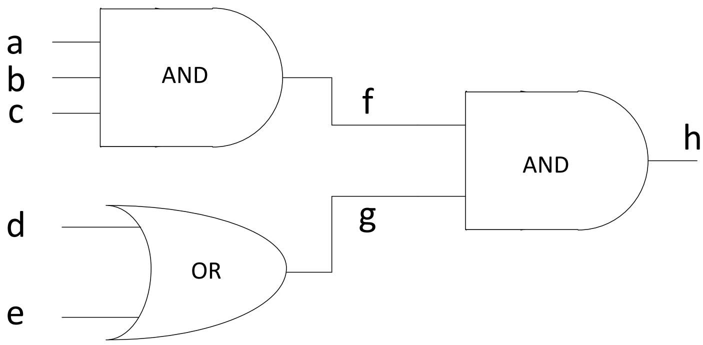
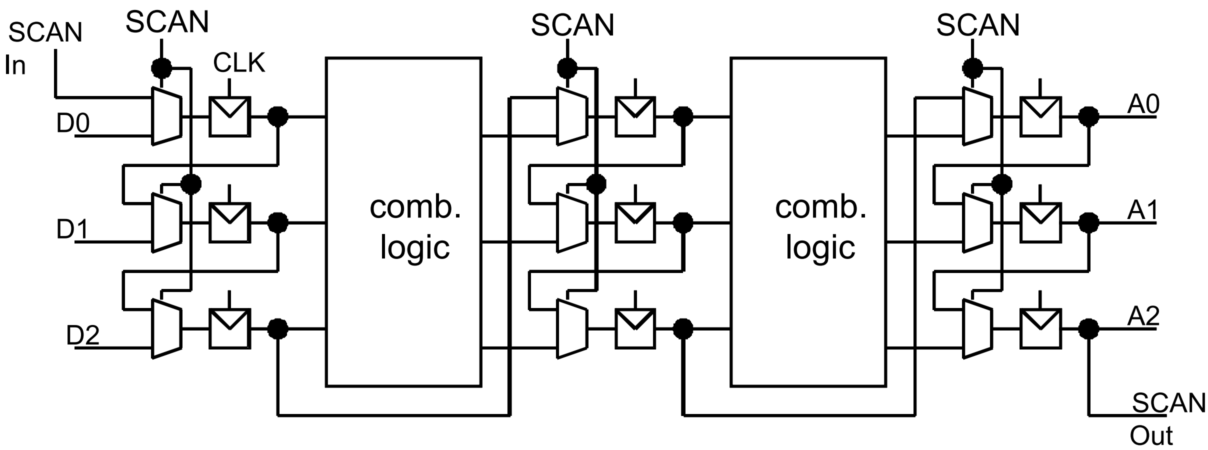
Testing: Test Pattern Generation, Scan-Design, BIST

1. The following circuit comes with five primary input and one primary output.

a) Determine all test vectors which can detect a possible stuck-at-0 fault on node f!
b) Determine all test vectors which can detect a possible stuck-at-1 fault on node f!

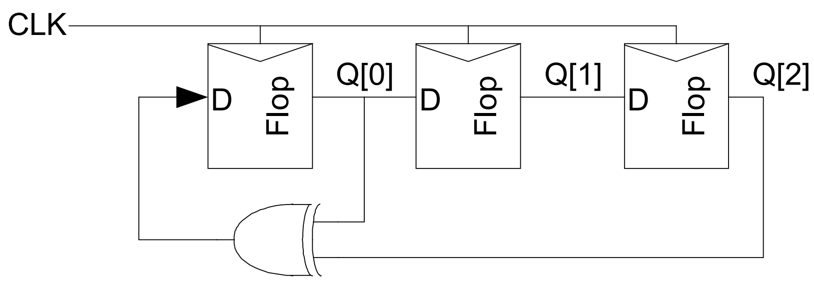


2. Explain the steps of testing the sequential circuit with scan path shown in the figure below!
   How many bits one test vector (pattern) needs? How many bits one test result have? How many clock cycles the test takes for one test vector?



3. Linear feedback shift registers (LFSR) are often used to generate test patterns. With an n-bit LFSR it is possible to generate 2^n-1 different bit patterns. All patterns can be generated except for all 0's! The patterns have certain randomness properties, so the LFSR works as a Pseudo Random Pattern Generator (PRSG).

Example: 3-bit LFSR



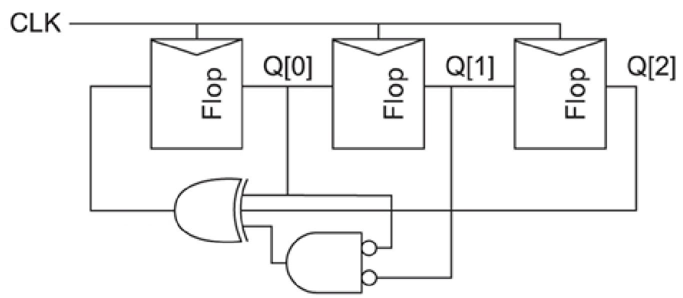
n	feedback
4,6,7	$Q_0 \oplus Q_{n-1}$
5	$Q_1 \oplus Q_4$
8	$Q_1 \oplus Q_2 \oplus Q_3 \oplus Q_7$
12	$Q_0 \oplus Q_3 \oplus Q_5 \oplus Q_{11}$
14,16	$Q_2 \oplus Q_3 \oplus Q_4 \oplus Q_{n-1}$
24	$Q_0 \oplus Q_1 \oplus Q_6 \oplus Q_{23}$
32	$Q_0 \oplus Q_1 \oplus Q_{21} \oplus Q_{31}$

Feedback for maximum length LFSR sequence

a) Draw a circuit schematic of a 4-bit LFSR that generates a maximum length sequence!
b) Determine the state sequence! Start with state $\{Q_3,Q_2,Q_1,Q_0\}$=1000!

4. If the all-0s test is required, an n-bit LFSR can be modified by adding an NOR-gate with n-1 inputs.

Example: 3-bit LFSR with NOR-gate



a) Determine the state sequence of this 3-bit LFSR with NOR-gate! Start with state 000!
b) Modify the 4-bit LFSR from 3.a) so that 0000 is included in the state sequence!